26900156 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECSC302 - ARTIFICIAL INTELLIGENCE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define state-space search. [U] / [CO1]
2. Define Knowledge Representation. [K] / [CO3]
3. Identify the connective represented by $P \wedge Q$. [U] / [CO3]
4. Briefly explain the reason the probability of the union of two mutually exclusive events A and B is equal to P(A) + P(B). [U] / [CO4]
5. State any two operations on classical/crisp sets. [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

6. Describe the major milestones in the early history of Artificial Intelligence. [U] / [CO1]

7. Explain how AI-based systems can improve agricultural productivity through crop monitoring, disease detection and resource management. [K] / [CO1]

8. Explain the relationship between expert knowledge and problem solving in an Expert System. [U] / [CO1]

9. Discuss how Expert Systems can support decision-making in Environmental Management. [A] / [CO1]

10. Explain the suitability of Generate and Test, Breadth-First Search, and Depth-First. [An] / [CO2]

11. Differentiate local heuristic and global heuristic using a suitable search-tree example. [An] / [CO2]

12. Explain how inheritance works in a category hierarchy and how it helps in knowledge inference. [U] / [CO3]

13. In a diagnostic test, Disease D has a prior probability P(D) = 0.01. The sensitivity of the test is 0.98, and the false positive rate is 0.05. If a patient tests positive, calculate the probability that the patient actually has Disease D. [A] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

14. Analyze the role of inference in Constraint Satisfaction Problems. Explain constraint propagation and arc consistency with suitable examples, demonstrate how inconsistent domain values are eliminated, and discuss how inference improves the efficiency of CSP search. [An] / [CO2]

15. Analyze the role of universal and existential quantifiers in First-Order Predicate Logic and illustrate their use in representing real-world knowledge. [An] / [CO3]

16. Analyze how linguistic variables and membership functions bridge the gap between human linguistic knowledge and numerical data. Using student performance as an example, explain how linguistic values such as Poor, Average, and Excellent can be represented using membership functions. Discuss how the resulting representation can handle gradual changes in the concept. [A] / [CO4]

[15x2 = 30]